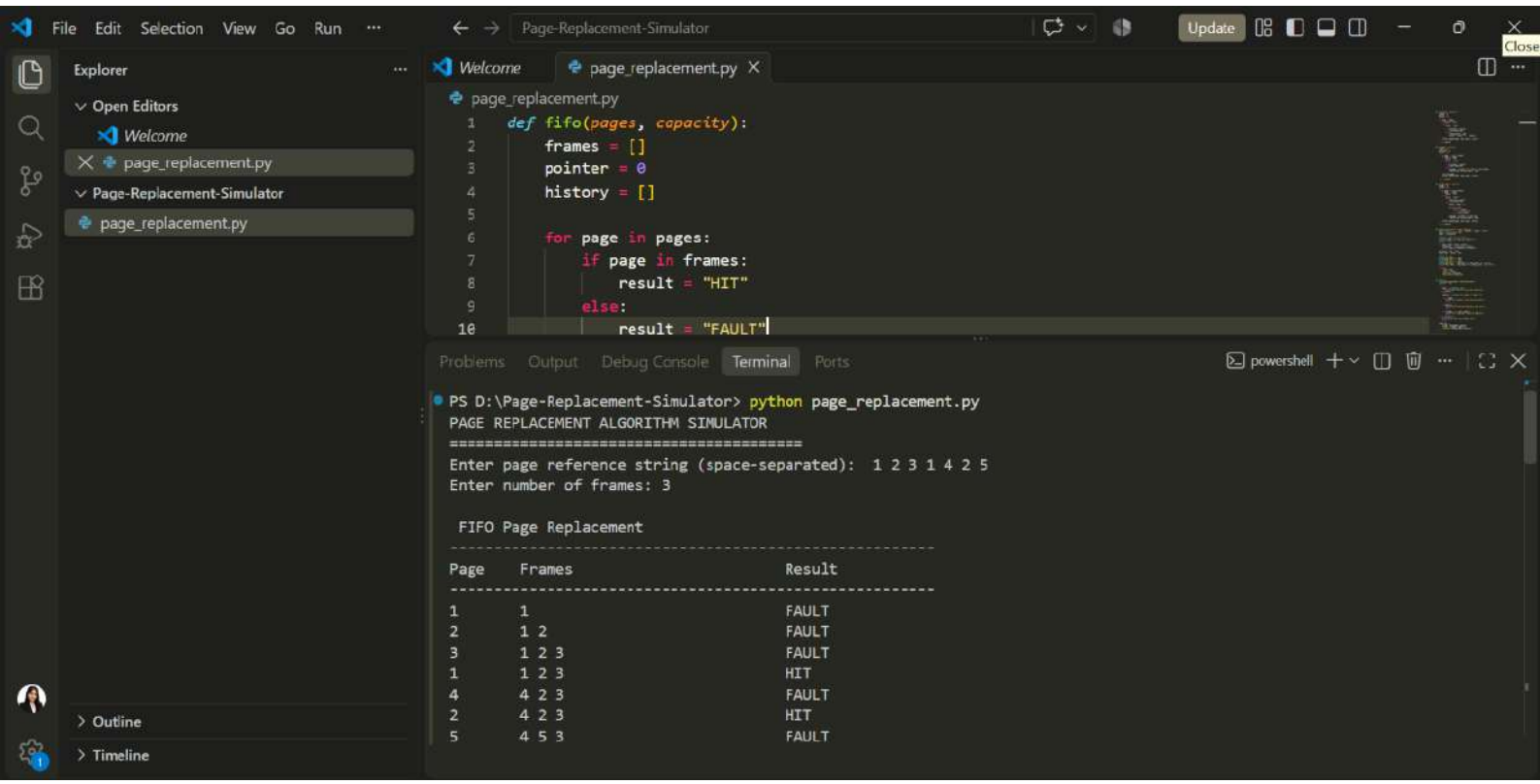
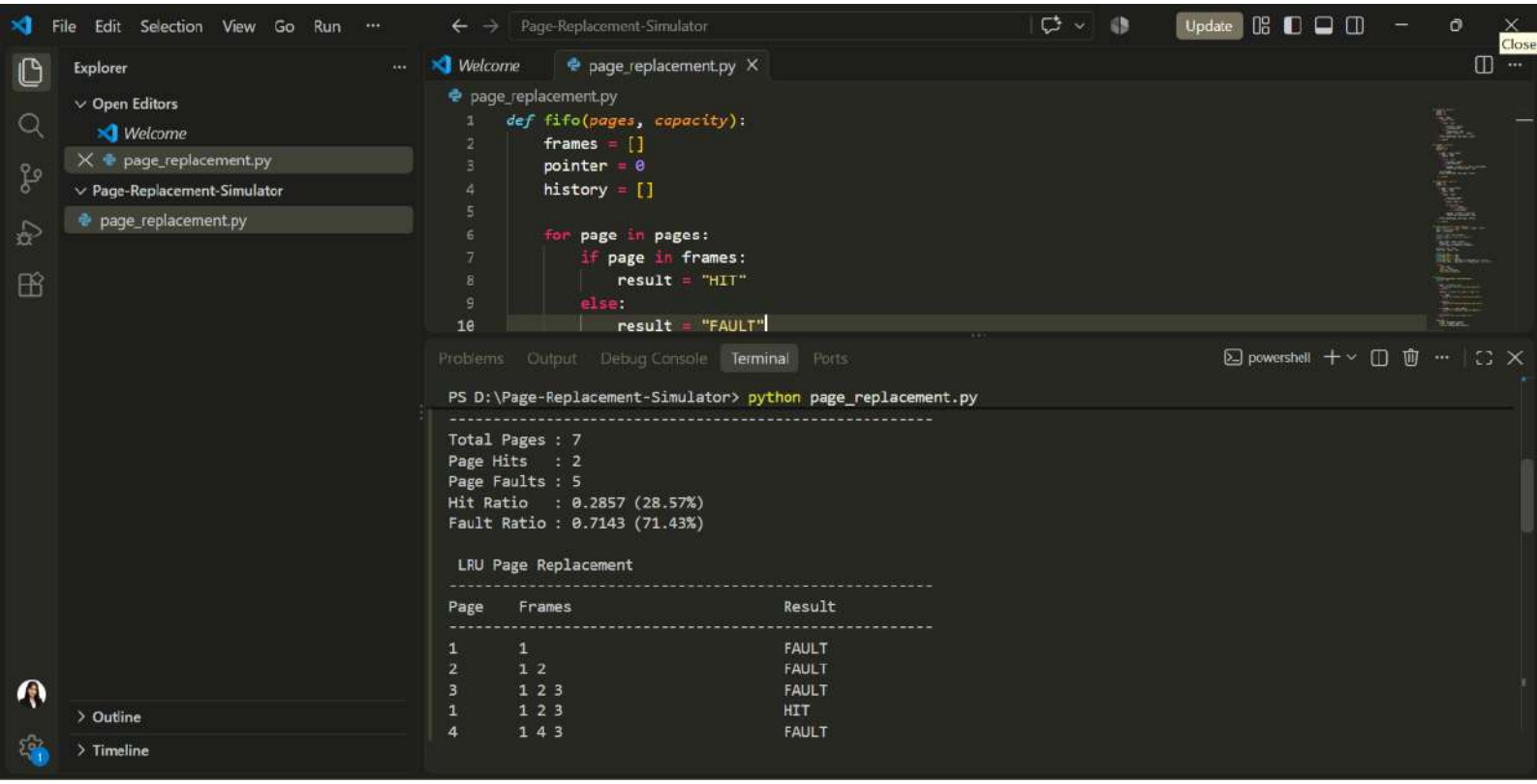
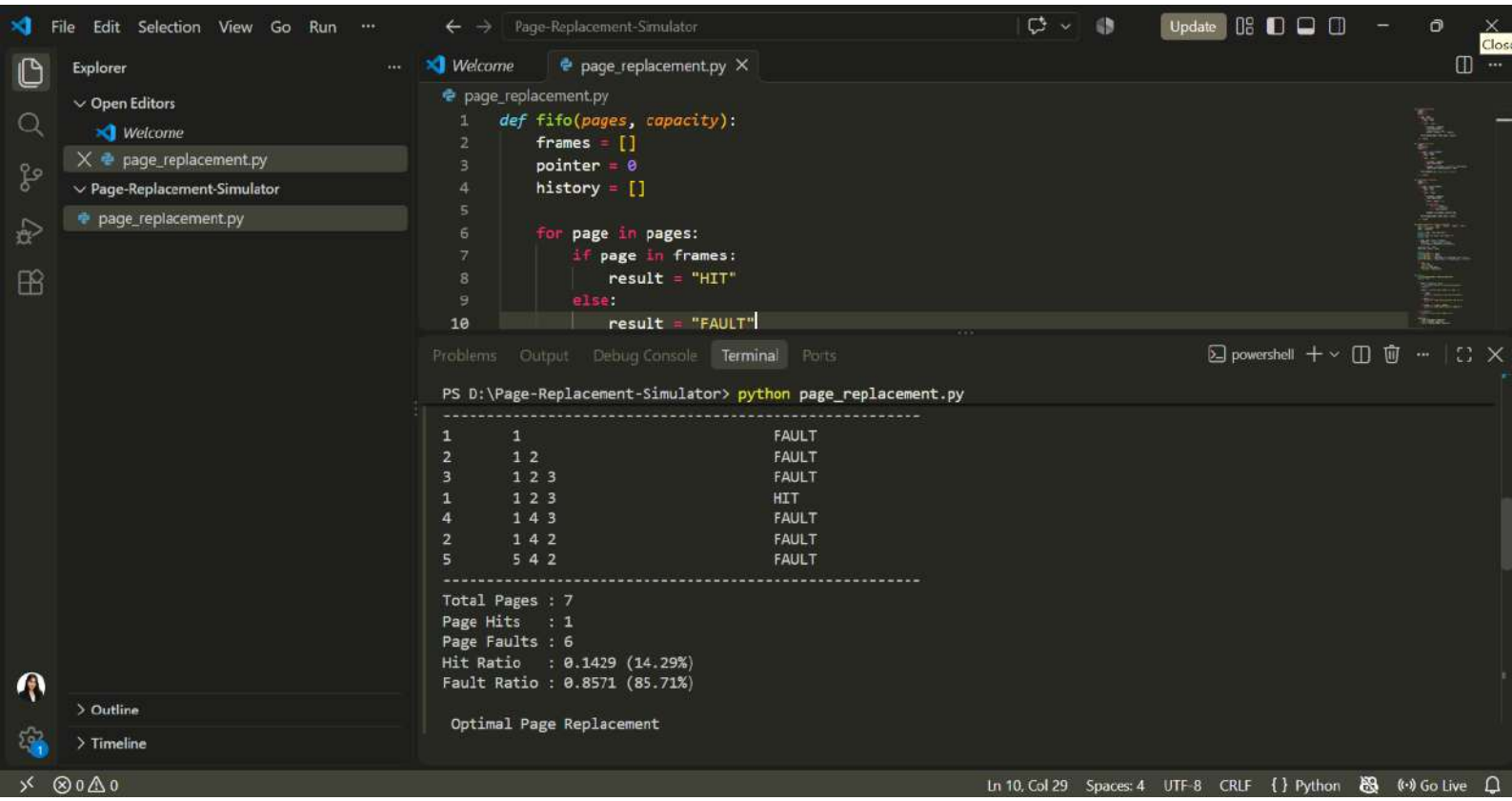
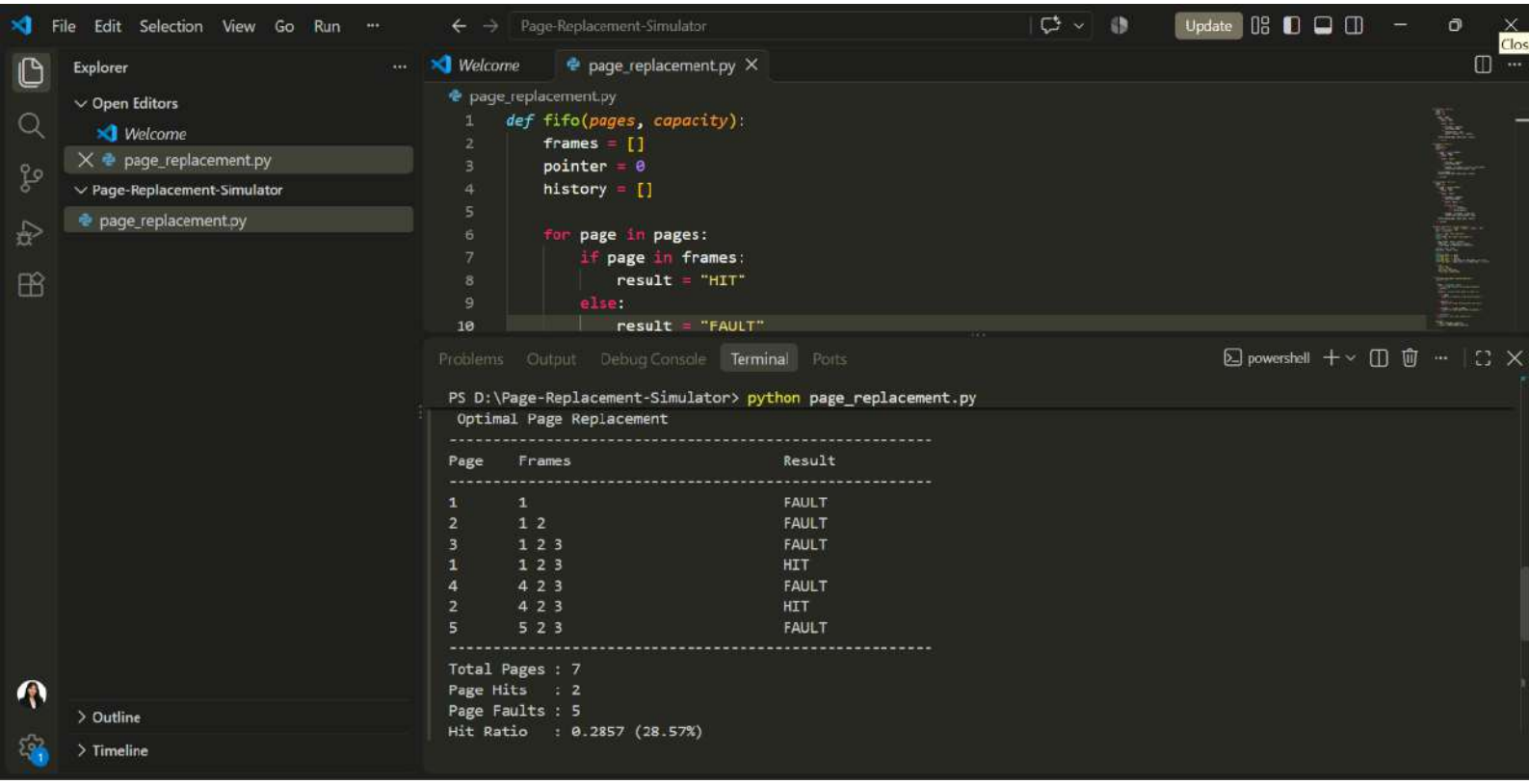
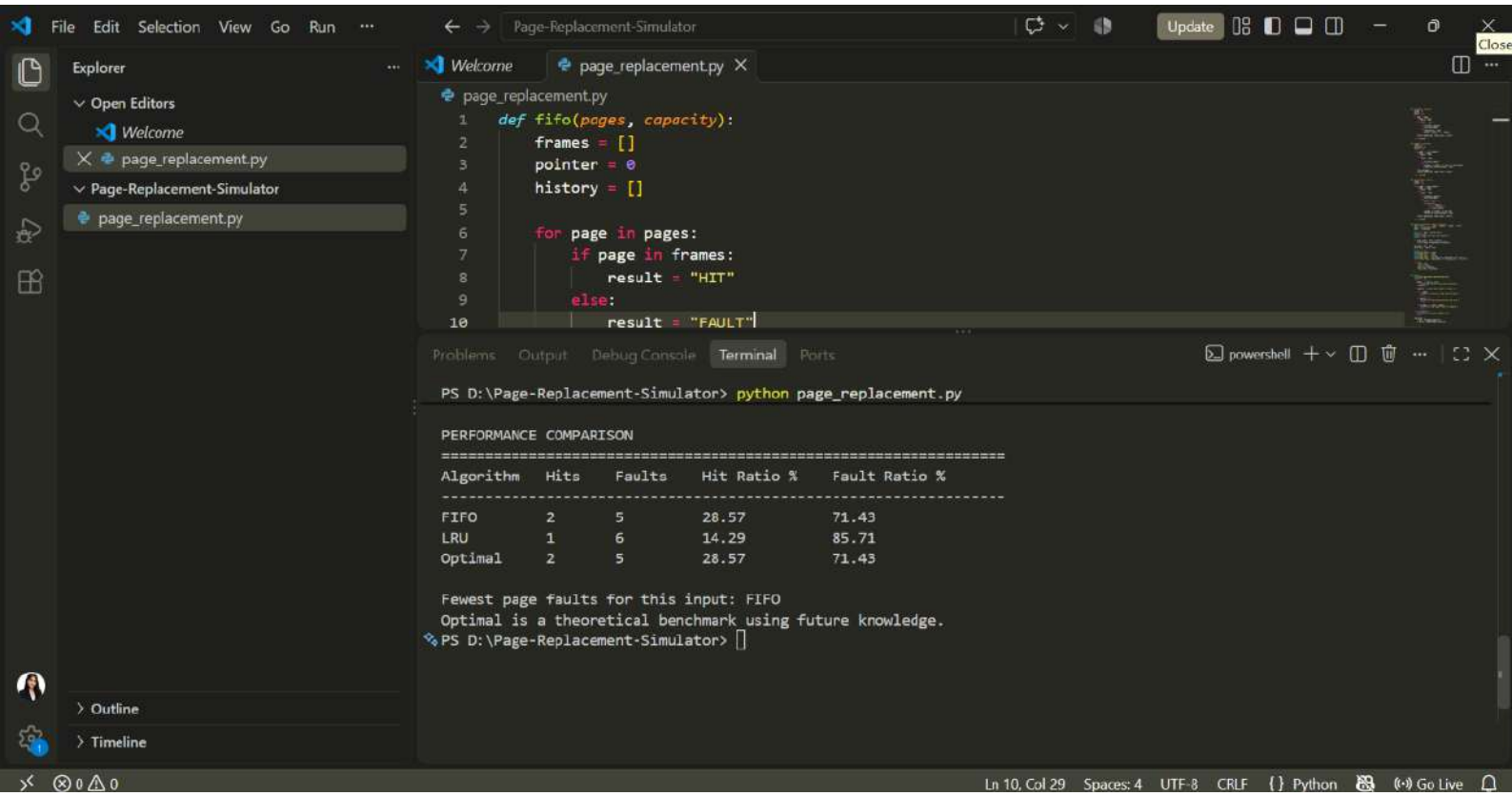

File Edit Selection View Go Run
Page-Replacement-Simulator
Update
Close
Explorer
Open Editors
Welcome
page_replacement.py
Page-Replacement-Simulator
page_replacement.py
Welcome
page_replacement.py
page_replacement.py
def fifo(pages, capacity):
frames = []
pointer = 0
history = []
for page in pages:
if page in frames:
result = "HIT"
else:
result = "FAULT"
Problems Output Debug Console Terminal Ports
powershell
PS D:\Page-Replacement-Simulator> python page_replacement.py
PAGE REPLACEMENT ALGORITHM SIMULATOR
Enter page reference string (space-separated): 1 2 3 1 4 2 5
Enter number of frames: 3
FIFO Page Replacement
Page Frames Result
1 1 FAULT
2 1 2 FAULT
3 1 2 3 FAULT
1 1 2 3 HIT
4 4 2 3 FAULT
2 4 2 3 HIT
5 4 5 3 FAULT
Outline
Timeline


File Edit Selection View Go Run
Page-Replacement-Simulator
Update
Close
Explorer
Open Editors
Welcome
page_replacement.py
Page-Replacement-Simulator
page_replacement.py
Outline
Timeline
Welcome
page_replacement.py
page_replacement.py
def fifo(pages, capacity):
frames = []
pointer = 0
history = []
for page in pages:
if page in frames:
result = "HIT"
else:
result = "FAULT"
Problems Output Debug Console Terminal Ports
powershell
PS D:\Page-Replacement-Simulator> python page_replacement.py
Total Pages : 7
Page Hits : 2
Page Faults : 5
Hit Ratio : 0.2857 (28.57%)
Fault Ratio : 0.7143 (71.43%)
LRU Page Replacement
Page Frames Result
1 1 FAULT
2 1 2 FAULT
3 1 2 3 FAULT
1 1 2 3 HIT
4 1 4 3 FAULT


File Edit Selection View Go Run
Page-Replacement-Simulator
Update
Explorer
Open Editors
Welcome
page_replacement.py
Page-Replacement-Simulator
page_replacement.py
Welcome
page_replacement.py
page_replacement.py
1 def fifo(pages, capacity):
2 frames = []
3 pointer = 0
4 history = []
5
6 for page in pages:
7 if page in frames:
8 result = "HIT"
9 else:
10 result = "FAULT"
Problems Output Debug Console Terminal Ports
powershell
PS D:\Page-Replacement-Simulator> python page_replacement.py
1 1 FAULT
2 1 2 FAULT
3 1 2 3 FAULT
1 1 2 3 HIT
4 1 4 3 FAULT
2 1 4 2 FAULT
5 5 4 2 FAULT
Total Pages : 7
Page Hits : 1
Page Faults : 6
Hit Ratio : 0.1429 (14.29%)
Fault Ratio : 0.8571 (85.71%)
Optimal Page Replacement
Outline
Timeline
Ln 10, Col 29 Spaces: 4 UTF-8 CRLF { } Python Go Live


File Edit Selection View Go Run
Page-Replacement-Simulator
Update
Explorer
Open Editors
Welcome
page_replacement.py
Page-Replacement-Simulator
page_replacement.py
Welcome
page_replacement.py
page_replacement.py
def fifo(pages, capacity):
frames = []
pointer = 0
history = []
for page in pages:
if page in frames:
result = "HIT"
else:
result = "FAULT"
Problems Output Debug Console Terminal Ports
powershell
PS D:\Page-Replacement-Simulator> python page_replacement.py
Optimal Page Replacement
Page Frames Result
1 1 FAULT
2 1 2 FAULT
3 1 2 3 FAULT
1 1 2 3 HIT
4 4 2 3 FAULT
2 4 2 3 HIT
5 5 2 3 FAULT
Total Pages : 7
Page Hits : 2
Page Faults : 5
Hit Ratio : 0.2857 (28.57%)
Outline
Timeline


File Edit Selection View Go Run
Page-Replacement-Simulator
Update
Close
Explorer
Open Editors
Welcome
page_replacement.py
Page-Replacement-Simulator
page_replacement.py
Welcome
page_replacement.py
page_replacement.py
1 def fifo(pages, capacity):
2 frames = []
3 pointer = 0
4 history = []
5
6 for page in pages:
7 if page in frames:
8 result = "HIT"
9 else:
10 result = "FAULT"
Problems Output Debug Console Terminal Ports
powershell
PS D:\Page-Replacement-Simulator> python page_replacement.py
PERFORMANCE COMPARISON
Algorithm Hits Faults Hit Ratio % Fault Ratio %
FIFO 2 5 28.57 71.43
LRU 1 6 14.29 85.71
Optimal 2 5 28.57 71.43
Fewest page faults for this input: FIFO
Optimal is a theoretical benchmark using future knowledge.
PS D:\Page-Replacement-Simulator>
Outline
Timeline
Ln 10, Col 29 Spaces: 4 UTF-8 CRLF Python Go Live